CASE STUDY 2: Attendance Calculation

Python Activity-Based Learning

1. Problem Statement

Write a Python program to calculate a student's attendance percentage and check whether the student is eligible based on a minimum attendance of 75%.

2. Objectives

• Read the total classes and classes attended.

• Calculate attendance percentage.

• Display eligibility.

3. Assumptions

• Total classes must be greater than zero.

• A minimum of 75% attendance is required.

4. Algorithm

1. Start.

2. Read total classes and attended classes.

3. Calculate percentage = attended / total × 100.

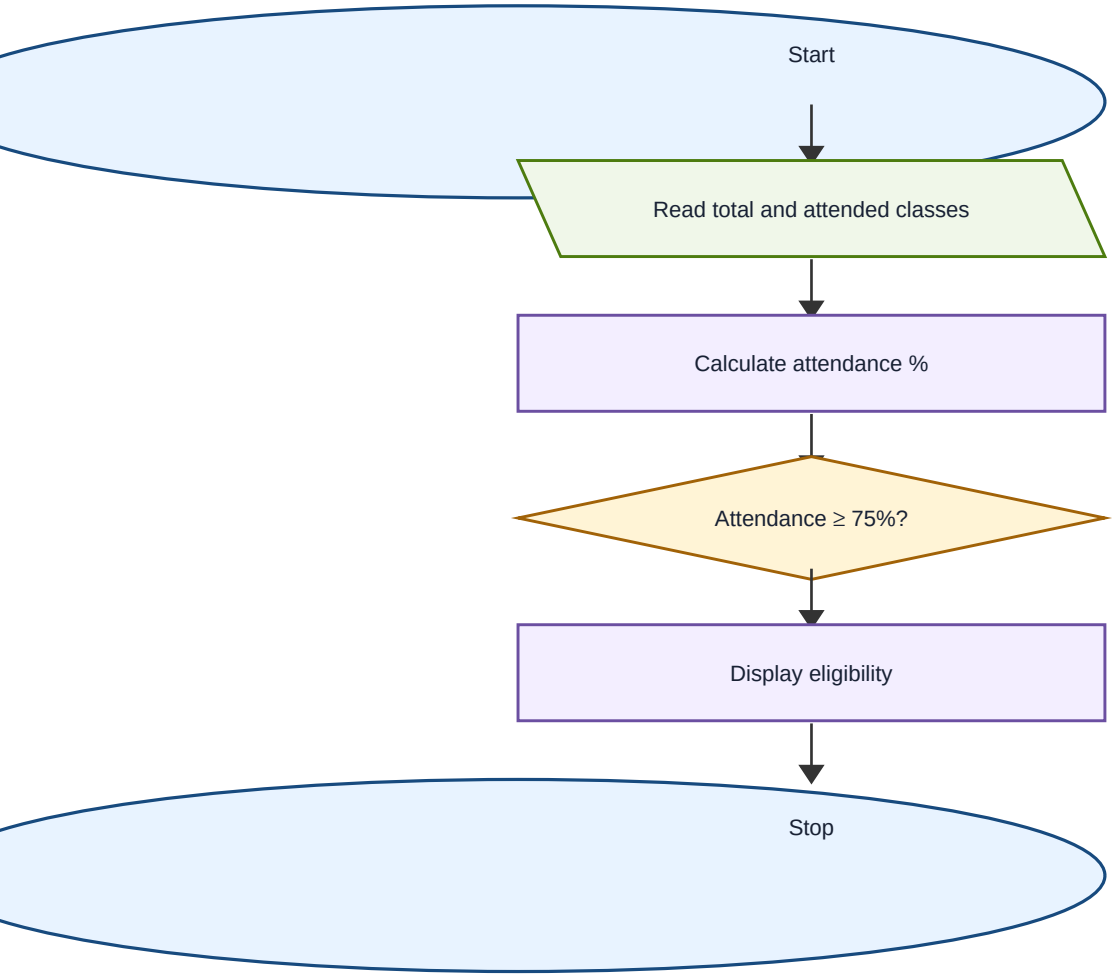
4. If percentage is at least 75, display Eligible; otherwise display Not Eligible.

5. Display the percentage.

6. Stop.

5. Flowchart

Oval = Start/Stop; parallelogram = Input; rectangle = Process; diamond = Decision.



6. Pseudocode

```
BEGIN
  INPUT total, attended
  percentage = attended / total * 100
  IF percentage >= 75 THEN
      PRINT "Eligible"
  ELSE
      PRINT "Not Eligible"
  END IF
  PRINT percentage
END
```

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7. Python Source Code

```
total = int(input("Enter total number of classes: "))
attended = int(input("Enter classes attended: "))

if total <= 0:
    print("Total classes must be greater than zero")
elif attended < 0 or attended > total:
    print("Enter valid attendance details")
else:
    percentage = (attended / total) * 100
    print("Attendance percentage:", percentage)

    if percentage >= 75:
        print("Status: Eligible")
    else:
        print("Status: Not Eligible")
```

8. Test Cases

Input	Expected Output
100 total, 80 attended	Attendance percentage: 80.0; Eligible
100 total, 60 attended	Attendance percentage: 60.0; Not Eligible

9. Sample Output

Run the Python program and enter the values from one of the test cases above. The displayed result should match the expected output.

10. Conclusion

The program uses arithmetic and if-else conditions to calculate attendance and check the 75% eligibility rule.